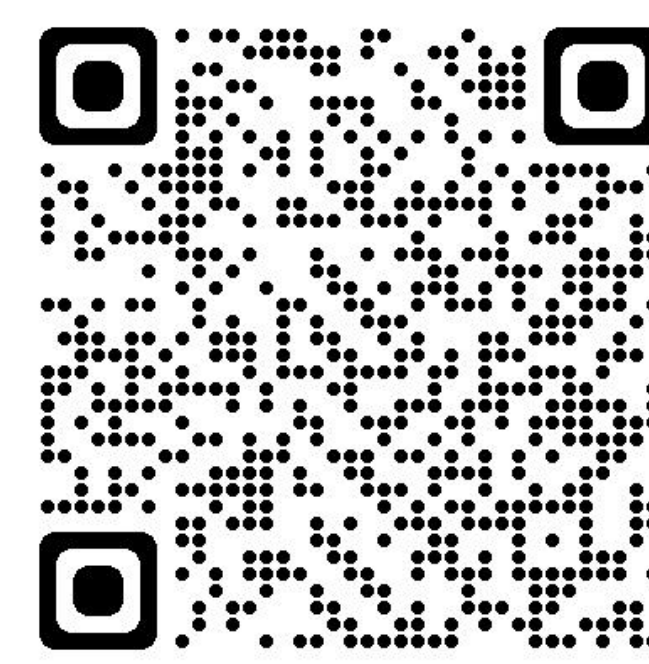


Predicting Video Slot Attention Queries from Random Slot-Feature Pairs

Rongzhen Zhao¹, Jian Li², Juho Kannala^{1,3}, Joni Pajarinen¹

¹EEA/CS, Aalto University; ²Renmin University of China; ³University of Oulu

source code,
model checkpoint
and training log:



What is Object-Centric Learning (OCL)

- discover objects in images or videos without supervision;
- represent them as corresponding feature vectors, slots.

Mainstream OCL on videos

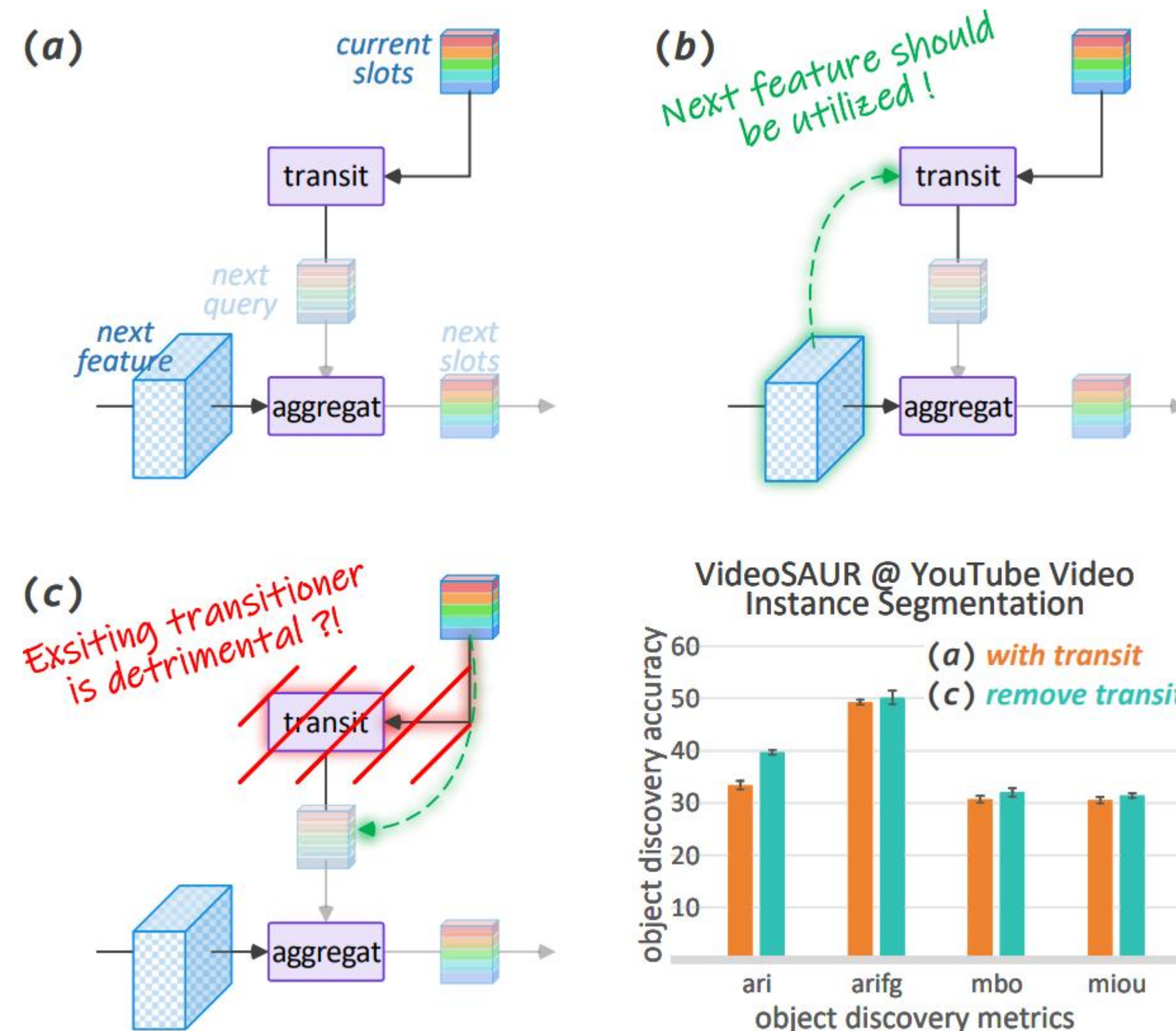
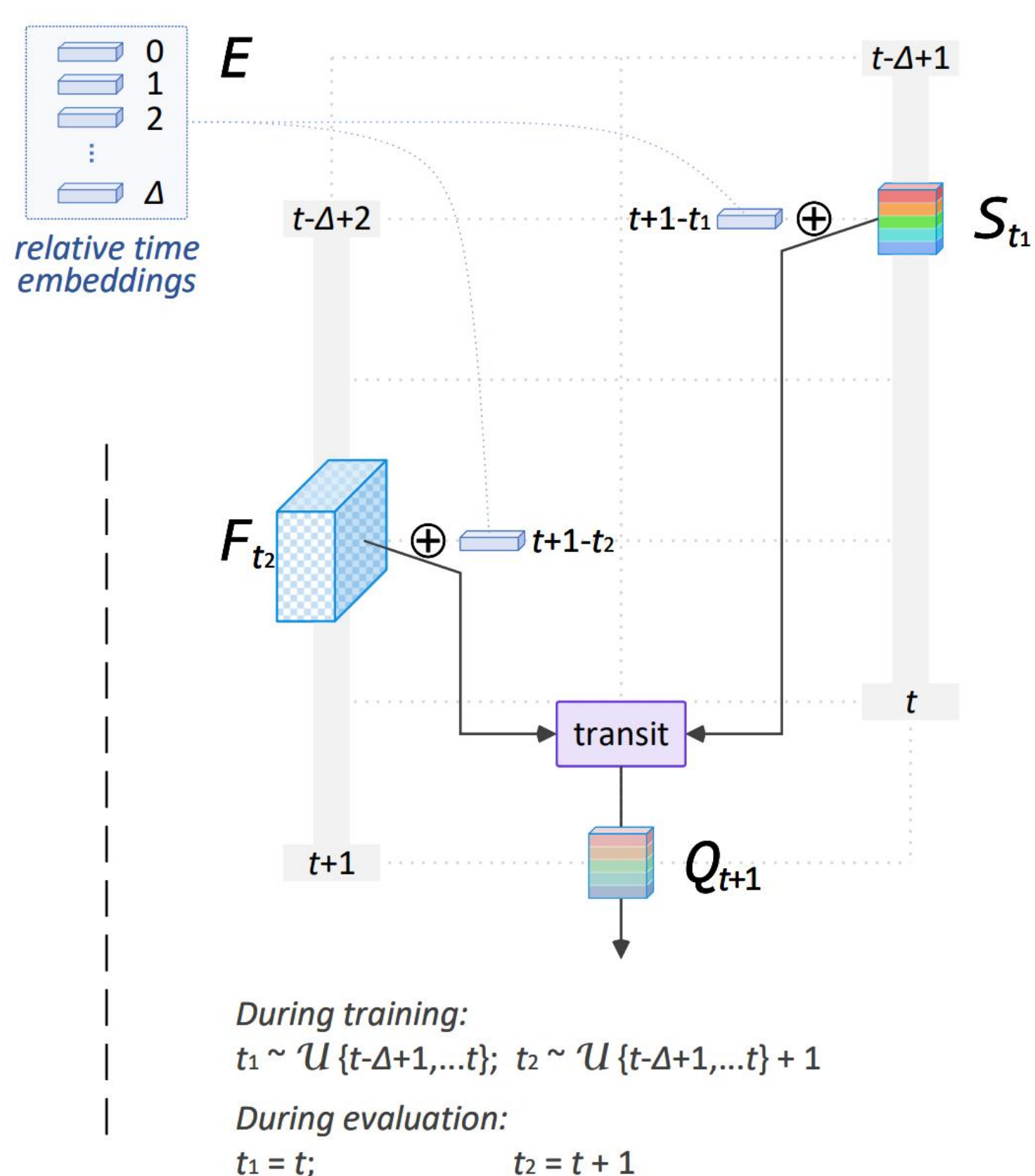
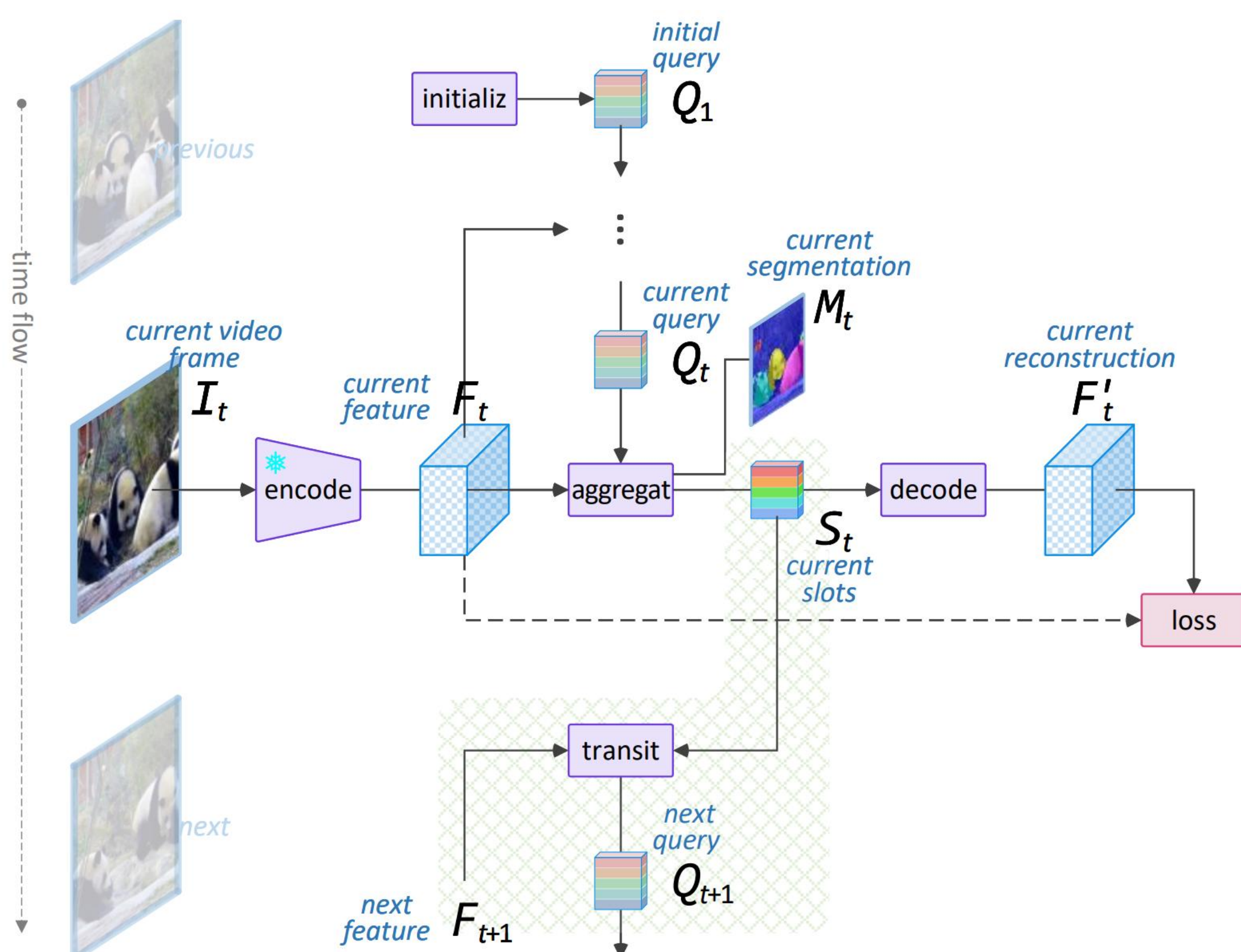
- adopt a recurrent architecture:
- an aggregator aggregates current video frame into object features / slots, under some queries;
- a transitioner transits current slots to queries for the next frame

How is an OCL model trained

- aggregate input's dense features into slots;
- reconstruct the input from such slots;
- minimize the reconstruction error.

Issues in mainstream video OCL

- neglect to incorporate next frame features, the most informative source for query prediction
- fail to learn transition dynamics, the knowledge essential for query prediction



Our solution RandSF.Q

- We design a new transitioner to incorporate both slots and features,
- which provides more information for query prediction;
- We train the transitioner to predict queries from slot-feature pairs randomly sampled from available recurrences,
- which drives it to learn transition dynamics

RandSF.Q boosts performance significantly

- scene representation / object discovery

	MOVi-C #slot=11, conditional				MOVi-D #slot=21, conditional				YTVIS #slot=7			
	ARI	ARI _{fg}	mBO	mIoU	ARI	ARI _{fg}	mBO	mIoU	ARI	ARI _{fg}	mBO	mIoU
STEVE	—	—	—	—	32.7 _{±0.2}	66.5 _{±0.2}	23.0 _{±0.3}	21.2 _{±0.3}	—	—	—	—
VideoSAUR	41.9 _{±1.1}	53.3 _{±2.1}	16.1 _{±0.4}	14.8 _{±0.4}	22.5 _{±5.0}	40.0 _{±20.1}	11.6 _{±6.6}	10.8 _{±6.1}	33.8 _{±0.7}	49.2 _{±0.5}	29.9 _{±0.4}	29.7 _{±0.4}
SlotContrast	64.6 _{±0.4}	59.9 _{±5.3}	27.7 _{±3.0}	25.8 _{±2.9}	45.3 _{±4.1}	63.9 _{±0.2}	26.7 _{±1.0}	25.1 _{±1.0}	37.2 _{±0.6}	49.4 _{±1.1}	33.0 _{±0.2}	32.8 _{±0.1}
RandSF.Q _{tsim}	64.0 _{±2.9}	66.3 _{±1.7}	28.4 _{±1.3}	26.1 _{±1.1}	41.2 _{±2.2}	72.0 _{±1.1}	27.1 _{±0.9}	25.4 _{±0.9}	46.0 _{±0.7}	60.4 _{±2.3}	39.4 _{±0.3}	38.5 _{±0.2}
RandSF.Q _{SSC}	65.4 _{±0.7}	67.4 _{±2.1}	29.2 _{±3.8}	26.8 _{±3.7}	41.6 _{±3.7}	77.5 _{±1.0}	27.4 _{±1.0}	25.6 _{±1.0}	40.1 _{±0.4}	58.0 _{±1.0}	37.6 _{±0.4}	37.2 _{±0.4}

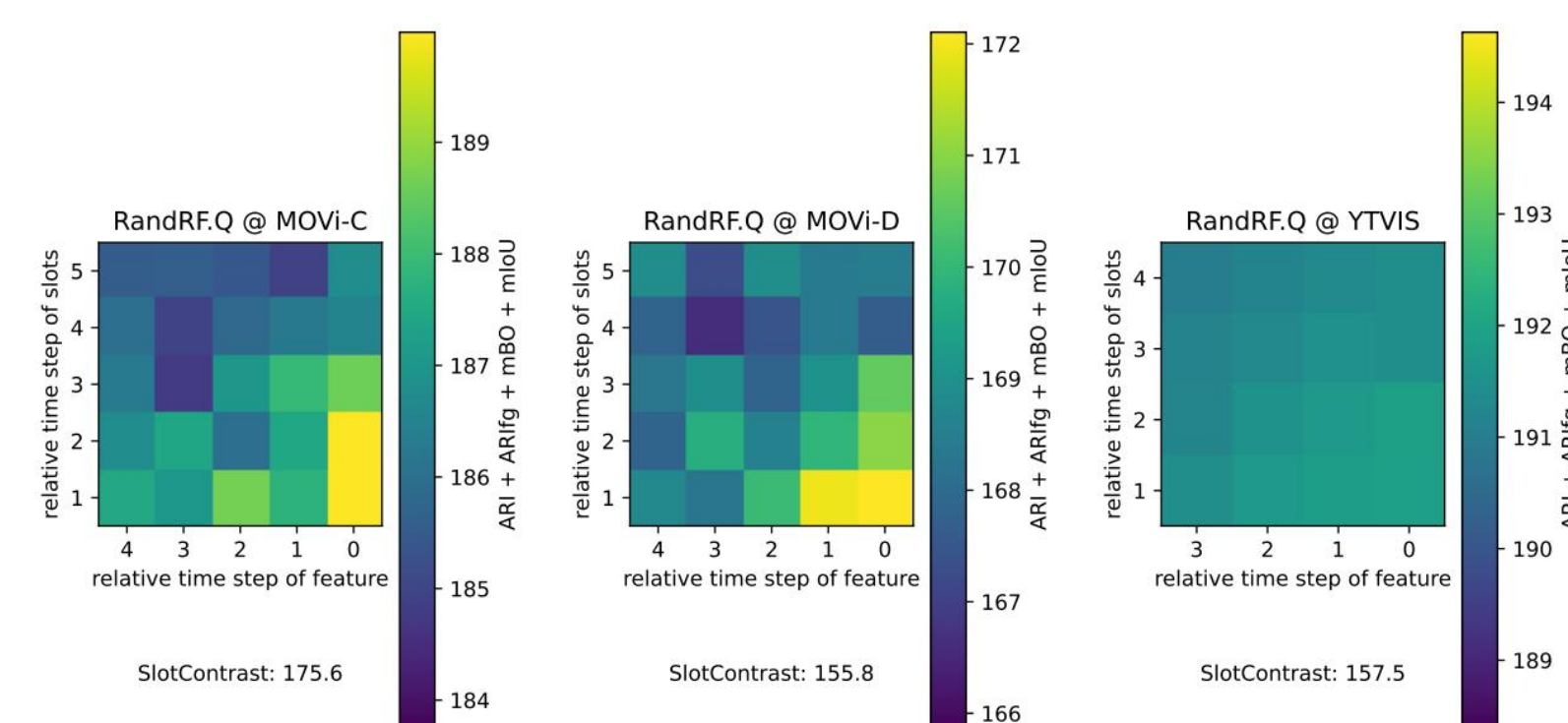
- scene representation / object recognition

	YTVIS #slot=7			
	class top1↑	top3↑	bbox IoU↑	#match↑
SlotContrast+MLP	19.9 _{±2.0}	49.1 _{±3.1}	53.5 _{±0.2}	9259 _{±26}
RandSF.Q +MLP	26.1 _{±1.3}	60.9 _{±3.2}	54.5 _{±0.6}	7579 _{±201}

- scene understanding / visual question answering

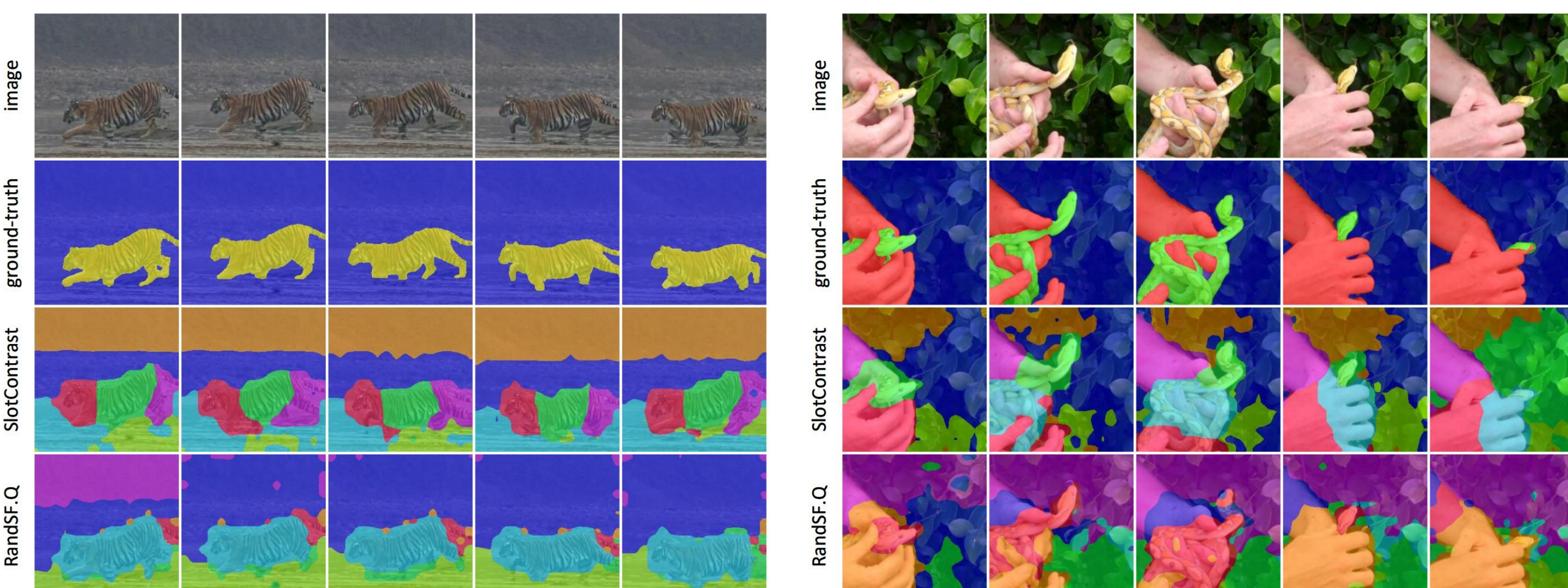
	CLEVRER #slot=7	
	per option %	per question %
SlotContrast+Aloe	97.2 _{±1.1}	95.6 _{±0.9}
RandSF.Q +Aloe	98.5 _{±0.8}	96.3 _{±0.7}

Implicit transition dynamics modeling



- RandSF.Q performance with queries predicted from slot-feature pairs at different relative time steps.
- The more up-to-date the slots and feature, the better the query prediction is and the better our model performs

Qualitative results



About me Rongzhen Zhao

- <https://generalz.github.io>
- 4th year Doctoral student
- looking for academic collaboration
- wrote the object-centric-bench framework adopted by many OCL works